Effects of Semantic Distance on Suppressed Concept Representations

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Seal or Signature

1. Summary of the Research

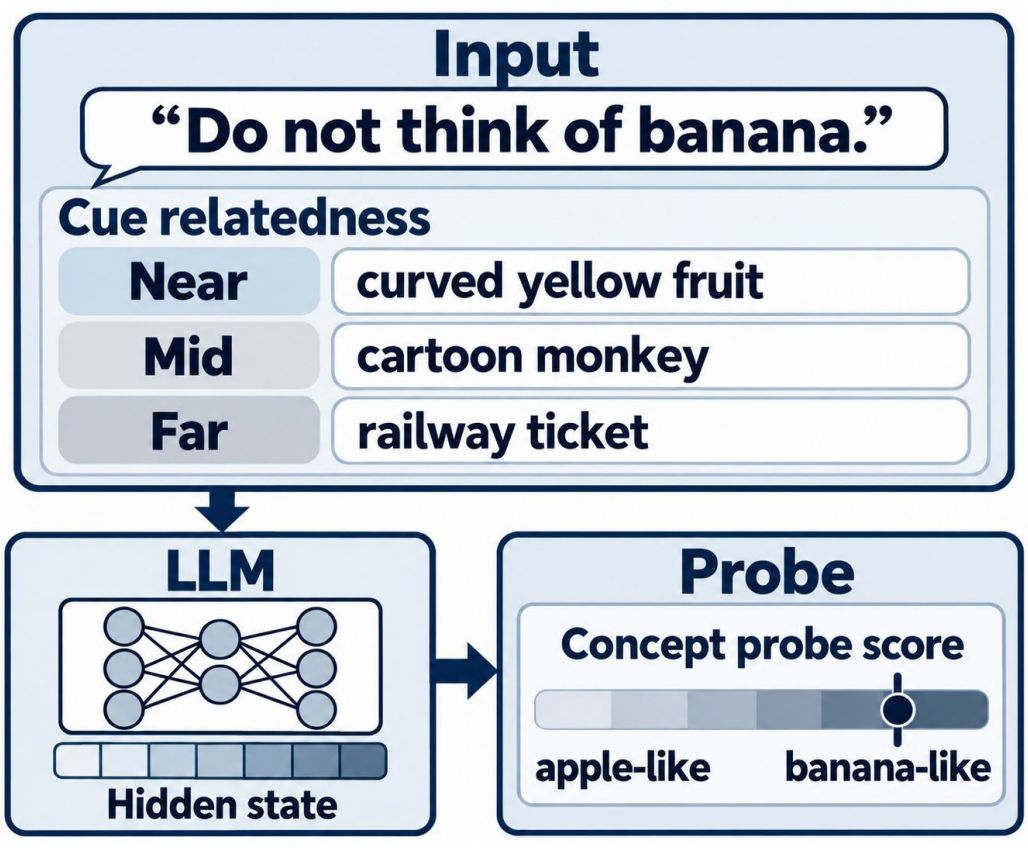
Large language models (LLMs) can be instructed not to think about a particular concept. However, avoiding a word in the final response does not necessarily mean that information about the concept disappears from the model's internal representations. Mann et al. [1] investigated thought suppression in LLMs under distracting contexts, while Zhou et al. [2] examined the internal mechanisms of negation. Building on these studies, we investigate how related information affects the representation of a suppressed concept in LLMs. We focus on how semantic distance between the cue and the target concept may influence its internal representation under different instruction conditions.

2. Approach / Methodology

We conduct experiments using Qwen2.5-0.5B-Instruct, a small language model. We test eight target concepts under four instruction conditions: neutral, affirmative, suppression, and replacement. After each instruction, we provide related or unrelated information and examine whether the target word appears in the generated response under each of these conditions.

We also extract hidden states from selected model layers and use a linear classifier (concept probe) to distinguish between target and comparison concepts. Fig. 1 illustrates our proposed experimental framework. We will introduce multiple levels of semantic distance and examine their effects on concept-probe scores derived from hidden states.

Fig. 1. Proposed Experimental Framework



3. Current Results and Status

Preliminary experiments were conducted using eight target concepts. We examined target-word mentions under four instruction conditions and two cue types, comparing their effects on generated responses across different settings. We also trained concept probes to examine the model's internal representations at selected layers. In a small validation set, the final-layer probe achieved a mean ROC-AUC of 0.972, indicating strong discrimination between target and comparison concepts based on hidden states.

Fig. 2 Target-Word Mention Rates

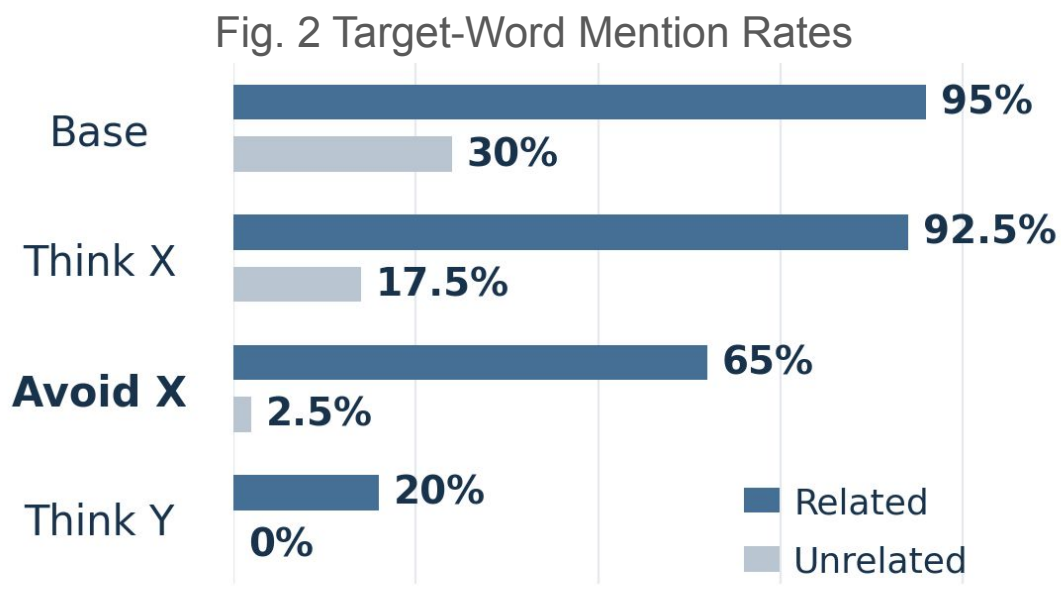


Fig. 2 shows that target-word mention rates were higher with related cues than with unrelated cues across all four instruction conditions. Under suppression, mention rates were 65% and 2.5%, respectively, showing a clear difference between cue types. However, suppression still reduced mentions compared with neutral instructions (95% with related cues). These preliminary results motivate further studies using graded semantic distances and hidden-state analysis to examine suppressed concept representations.

4. Remaining Tasks and Schedule

- **Oct** Prepare semantic-distance cues
- **Nov** Analyze hidden state effects
- **Dec** Compare experimental conditions
- **Jan** Summarize results & write thesis

References

[1] L. Mann et al., “Don't Think of the White Bear: Ironic Negation in Transformer Models Under Cognitive Load,” *arXiv:2511.12381*, 2025.

[2] Z. Zhou et al., “How Language Models Process Negation,” *Proc. ICML*, PMLR 306, pp. 166474–166494, 2026.